Muon Against SGD and AdamW, and Late-Stage Spectral Shaping: Behaviour Across Batch Size and Momentum

Chee Jia Yun (A0272291E)
Clarice Ng Kai Li (A0277515X)
Tasha Lim Shi Ying (A0264170L)

Code, configs and results:
`https://github.com/jiayunchee/dsa4212_Muon`

Abstract

We study late-stage spectral shaping in Muon on a synthetic teacher–student regression task, replacing its update with $US^pV^\top$, where $p = 0$ is exact Muon and $p < 0$ gives weak directions more weight. With learning rates tuned separately for each optimiser, Muon reaches lower tail excess risk than SGD and AdamW after 6000 steps at all four batch sizes, but per estimated CPU compute time only NS5 Muon is ahead, and only at the largest batch size. In paired branches from shared checkpoints, evaluated on independent seeds, $p = -0.25$ lowers mean tail excess risk by approximately 2.4–7.6% relative to continued exact Muon in all eight batch-size and momentum conditions; SGD-like updates do worse, and the planned noise hypotheses are not supported. An exploratory extension favours $p = -0.5$ at larger batch sizes, and an initially incomplete tuning search shows how strongly such comparisons depend on learning-rate selection. Results come from one synthetic task, three evaluation seeds per branch condition and baselines reported on their tuning seeds, and do not isolate gradient noise as a cause.

1 Introduction

Muon transforms the momentum updates of matrix-valued hidden weights. Its practical version uses a short Newton–Schulz iteration to approximately flatten the spectrum (Jordan et al., 2024). Increasing the relative weight of weak directions may help optimisation, but may also amplify gradient noise. Xie (2026) analyse a noise-dominated regime where response-matched momentum can avoid a nonlinear residual introduced by orthogonalisation. Wu et al. (2026) instead motivate mildly negative powers late in training. These arguments address different settings and do not determine the outcome on our task.

We investigate whether modifying Muon's spectral update improves late-stage optimisation on a synthetic teacher–student regression task. We first compare Muon with SGD and AdamW under fixed training-step budgets and estimated computation-time budgets. We then compare separately tuned spectral exponents from shared checkpoints across batch sizes and momentum settings. A gradient-noise indicator measured at the switch helps us explore whether the observed differences are associated with noise. Exploratory controls and additional exponents investigate the sensitivity of the results to step-size selection.

Our contributions are as follows.

- A comparison of Muon with SGD and AdamW under comparable tuning effort, reported both per training step and per estimated compute time. Muon has lower excess risk per

step at all four batch sizes, but per unit of compute only NS5 Muon beats SGD, and only at the largest batch size.

- A planned branch experiment across eight batch-size and momentum conditions. Flattening more than Muon ($p = -0.25$) improves on continued Muon in every condition, while the SGD-like update and Xie's switch increase excess risk.
- Exploratory evidence that the preferred exponent shifts to $p = -0.5$ at larger batch sizes, and a documented case in which an incomplete learning-rate search changed the apparent ranking of update shapes.

The initial experiment protocol specified three hypotheses:

1. **H1: Preferred exponent.** Higher gradient noise favours larger late-phase spectral exponents. We therefore expected larger exponents to perform better at smaller batch sizes or lower momentum, and non-positive exponents to be preferred at larger batch sizes or higher momentum.
2. **H2: Switching to momentum SGD.** Switching from Muon to an unnormalised momentum update, using Xie's matched learning-rate scale, reduces late-stage excess risk. We expected a larger benefit at smaller batch sizes and lower momentum.
3. **H3: Noise indicator.** The signal-to-noise indicator δ measured at the switch predicts the preferred exponent: smaller δ, indicating more noise relative to signal, favours larger exponents.

These hypotheses motivated the original branch experiment. The additional negative exponents and frozen-hidden-layer control were introduced after inspecting earlier results and are treated as exploratory.

2 Background

For a hidden weight matrix $W \in \mathbb{R}^{m \times n}$ with Nesterov momentum input $N = USV^\top$ (thin SVD, numerical rank r), we study the update family

$$D_p = \frac{\sqrt{r}}{\|US^pV^\top\|_F}\, US^pV^\top, \qquad W \leftarrow W - \eta\sqrt{\max(1, m/n)}\, D_p. \tag{1}$$

The exponent p controls how the update distributes its strength across the singular directions of the momentum input N. Before transformation, a larger singular value indicates a direction with a stronger contribution to N. For the retained singular values, $p = 1$ preserves their relative magnitudes, producing a Frobenius-normalised momentum-like update. Setting $p = 0$ makes them equal, giving the exact polar update used by idealised Muon. A negative exponent gives relatively greater weight to directions with smaller singular values. These weak directions may contain useful remaining signal, but they may also contain noise.

To separate spectral shape from overall update magnitude, we normalise each direction D_p to Frobenius norm $\sqrt{r}$ and tune the learning rate separately for each arm. Before applying the power, we discard singular values at or below $s_1 \max(m, n)\epsilon$, where ϵ is float32 machine precision and r is the number retained. This prevents negative powers from strongly amplifying numerically negligible singular values.

The normalised $p = 1$ update differs from the separate Xie-style arm, which uses an unnormalised momentum direction with a matched learning-rate scale. Practical NS5 uses five Newton–Schulz iterations to approximate the polar transformation; its resulting spectrum differs from the exact-SVD update (Jordan et al., 2024).

Xie's analysis motivates the signal-to-noise indicator δ and a matched switch (Xie, 2026); DynMuon schedules the exponent across training (Wu et al., 2026). Our fixed-exponent branches do not implement the complete DynMuon schedule. Spectral shape and step size can be confounded, motivating separate tuning for each arm (Shumaylov et al., 2026).

Related work. Bernstein and Newhouse (2024) interpret several optimisers as steepest descent under a chosen norm: under the spectral norm the step is the orthogonalised matrix $UV^\top$, and under the Frobenius norm it is the normalised gradient. Applied to the momentum input, these are our $p = 0$ and $p = 1$ updates, so the family in (1) interpolates between the two geometries and, for $p < 0$, extends beyond the spectral end. Liu et al. (2025) identify weight decay and adjusting the per-parameter update scale, matching the update's root-mean-square size to AdamW's, as crucial for scaling Muon to large language models. Our runs use zero weight decay, and because our two hidden matrices share one shape, a shape-dependent rescaling would only change the overall step size, which we tune. Other variants reshape the update in ways we do not test: NorMuon normalises each neuron's row of the orthogonalised update using second-moment statistics (Li et al., 2025), and Dion orthonormalises a low-rank approximation of the momentum for sharded distributed training, carrying the remainder forward through error feedback (Ahn et al., 2025).

3 Setup and protocol

We study synthetic teacher–student regression because it provides a known clean prediction target and reproducible streams of noisy training examples. Unequal input variances let us compare update rules in a problem with differently scaled input directions, while a small network keeps repeated tuning and controlled branch comparisons feasible on a CPU. Inputs are sampled from a 128-dimensional Gaussian distribution with covariance eigenvalues proportional to $1/i$, rescaled so that their sum is 128. A fixed randomly initialised teacher network has widths 128–64–64–16, tanh hidden activations, and a linear output layer. Its outputs are standardised using estimated per-output means and standard deviations. Training labels add independent Gaussian noise with standard deviation 0.5 to these outputs, and a fresh mini-batch is generated at every training step.

The student has widths 128–256–256–256–16, with tanh hidden activations and a linear output layer. In Muon-based runs, only the two 256×256 hidden weight matrices receive Muon or its variants. The input and output weight matrices and all biases use AdamW (Loshchilov and Hutter, 2019). The standalone AdamW and SGD baselines apply their respective optimiser to every student parameter. AdamW uses $(\beta_1, \beta_2) = (0.9, 0.999)$. In Muon-based runs, its learning rate for the remaining parameters is the rate selected for the standalone AdamW baseline at the same batch size. Weight decay is zero in all experiments.

We evaluate predictions on 16,384 fixed inputs using noiseless teacher outputs as targets. The mean squared prediction difference, averaged over inputs and output coordinates, estimates excess risk above the Bayes predictor. Using noiseless targets avoids having to estimate and subtract the label-noise floor. Our main outcome is tail excess risk: the average of evaluation measurements in the final 10% of each training run or continuation branch.

In E1, each optimiser is trained for 6000 steps at each batch size, starting with a grid of five learning rates. SGD uses Nesterov momentum 0.9, while both Muon variants use Nesterov momentum 0.95; these momentum settings are fixed during E1 tuning. The SGD grid is extended at $B = 1024$. Rates are selected using mean tail excess risk over seeds 0 and 1, and the reported E1 outcomes use those same seeds. These results therefore describe performance after tuning on the reported seeds, rather than an independent evaluation of the selected settings.

For the branch experiments, E2 trains exact Muon for 3000 steps at each combination of batch size $B \in \{16, 64, 256, 1024\}$ and momentum $\beta \in \{0.9, 0.95\}$. Each E3 or F1 branch then continues for a further 3000 steps. Within each condition and seed, branches share the starting model, optimiser state, and subsequent training-data stream. The hidden-matrix update rule and learning rate may change at the switch, while the AdamW state and learning rate for the remaining parameters are preserved. Learning rates remain constant within each phase.

Each branch arm is tuned on seeds 0 and 1 and evaluated on seeds 2, 3 and 4. Tuning starts with learning-rate multipliers $\{0.5, 1, 2\}$ and extends the grid when the best value lies at an edge, subject to the documented bounds. Consequently, tuning trial counts differ across arms and conditions. Selected multipliers are reported in Table 3. We compare branches using paired differences over the three evaluation seeds; the resulting t-intervals should be interpreted cautiously because the sample is small.

The E3 question, initial arms and comparisons, and the requirement to extend any tuning grid whose best value lies at an edge, were specified in the protocol before the E3 runs; the exact extension procedure and the bounded stopping rule were fixed during tuning. The frozen control and F1 were added after inspection and are exploratory. The timing procedure was changed to five-pass medians after a noisy single pass. The dated change log records these decisions, and Table 1 summarises all experiments, their run counts and their status. The per-layer indicator $\delta = \|\widehat{EG}\|_F/(\widehat{\sigma}_N\sqrt{mn})$ uses 32 fresh minibatch gradients, bias correction of the squared mean-gradient norm, and a momentum-noise calculation assuming stationarity and independent gradient noise. We average the estimate across both hidden matrices and seeds.

Table 1: Experiment overview. E3 and F1 use separate tuning and evaluation seeds; E1 reports results on its tuning seeds. Training runs use CPU computation in float32.

	Purpose	Runs	Seeds	Status
E0	Pilot: timing and learning-rate ranges	16	0	pilot
E1	Baselines, learning rate tuned per optimiser and batch size	164	0, 1	planned
E2	Shared exact-Muon phase (3000 steps), checkpoints	40	0–4	planned
E3	Branches $p \in \{-0.25, 0, 0.5, 1\}$, NS5, Xie: tuning / evaluation	470 / 144	0, 1 / 2–4	planned[c]
E3c	Branches with the hidden layers frozen	40	0–4	exploratory
F1	Branches $p \in \{-0.75, -0.5\}$: tuning / evaluation	212 / 48[a]	0, 1 / 2–4	exploratory
E-T	Time per step, median of 5 passes	–	–	amended[b]

[a]Plus 45 superseded runs at a step size selected before tuning had finished; kept, not used. [b]Changed from one pass to the median of 5 passes after single passes proved noisy. [c]Tuning and stopping rules were amended during tuning; see the protocol change log.

4 Muon against the baselines, per step and per second

At 6000 steps, exact Muon has approximately 37.5, 25.2, 29.1 and 45.1% lower tail excess risk than the better baseline, SGD, at $B = 16, 64, 256, 1024$, respectively; NS5 is close to exact Muon (Figure 1).

For the equal-time comparison, we estimate training-compute time by multiplying the number of steps by a separately benchmarked time per step. The benchmark measures the forward pass, backward pass, and optimiser update on one CPU thread; it excludes data generation, evaluation, and setup. We use the median step time across five benchmark passes and interpo-

late the seed-averaged E1 learning curves at the compute budget corresponding to 6000 SGD steps.

This comparison retains learning rates selected for the 6000-step outcome rather than retuning them for the equal-time budget. It therefore compares the selected configurations under an estimated compute budget, rather than establishing each optimiser's best achievable performance at that budget. SGD has lower estimated risk than both Muon variants at $B \leq 256$, whereas NS5 Muon has lower estimated risk at $B = 1024$ (about 0.0058 versus 0.0079). Exact-SVD Muon is within 2.4% of NS5 per step but takes 2.0–2.7 times as long per step, and at equal time it has the highest estimated risk of the four optimisers at every batch size; on this task, the more accurate orthogonalisation does not repay its cost. These results depend on the tested CPU and benchmark conditions. The pilot's different ranking also illustrates the importance of tuning before comparing methods.

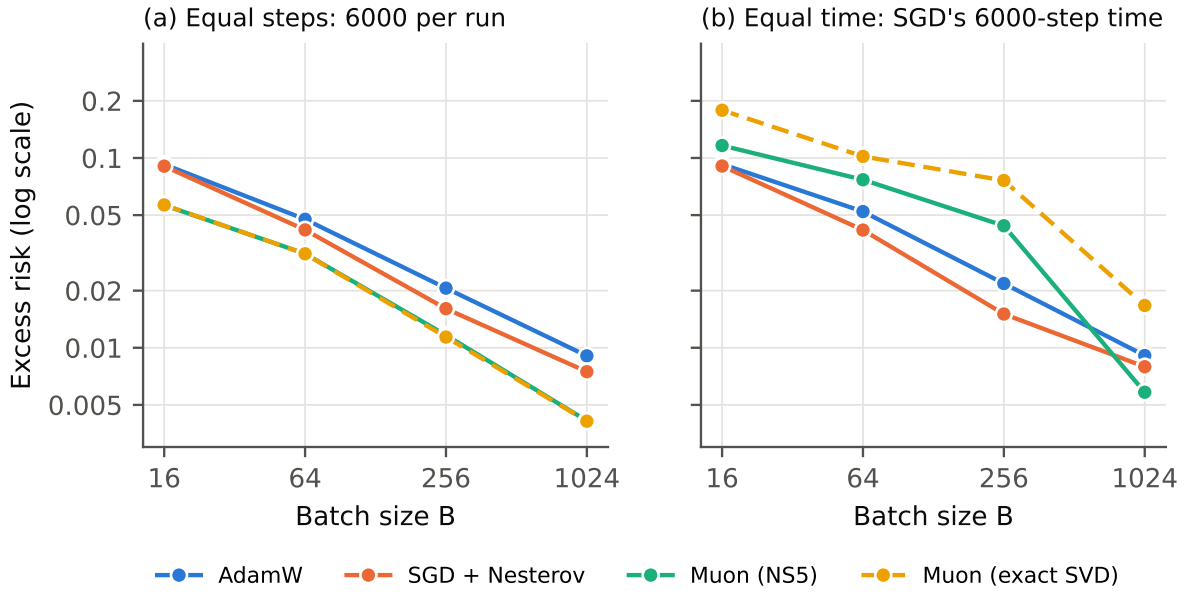


Figure 1: Tuned baselines on the synthetic teacher–student task. (a) Tail excess risk, averaged over the final 10% of 6000 training steps and two seeds, at each optimiser's selected learning rate. (b) Estimated excess risk at the training-compute budget corresponding to 6000 SGD steps, obtained by interpolating seed-averaged learning curves using separately benchmarked step times. Timing includes forward and backward passes and optimiser updates, but excludes data generation, evaluation, and setup. Learning rates are selected for the fixed-step outcome in both panels. Muon achieves lower tail risk per step at every batch size; under the estimated equal-time budget, only NS5 Muon outperforms SGD, and only at $B = 1024$.

5 Original branch experiment: the late-phase exponent under noise

Table 2 and Figure 2 show the paired evaluation. Relative to continuing exact Muon, $p = -0.25$ improves tail risk by approximately 2.4–7.6% across conditions, while $p = 1$ and the Xie-style switch increase it by about 12–41% and 12–37%, respectively. NS5 remains close to exact Muon. The best tested exponent in the original E3 grid is -0.25 in all eight conditions.

The planned H1 contrast compares the absolute risk gap between $p = 1$ and $p = -0.25$ at $B = 16$ with the corresponding gap at $B = 1024$. H1 predicts a negative contrast. Instead, the analysis gives $C = 0.00877$ [0.00833, 0.00922] for $\beta = 0.9$ and 0.00657 [0.00617, 0.00698]

for $\beta = 0.95$ (Welch 95% intervals; differences between arms within each batch size are paired over seeds). By the pre-specified criterion, H1 is therefore contradicted: both intervals lie above zero, and at $B = 16$ the SGD-like exponent $p = 1$ is 12–16% worse than continuing Muon rather than better. However, this contrast measures absolute gaps between conditions with substantially different risk levels, so its sign does not by itself establish how noise changes the relative advantage of one exponent over another. Whether the preferred exponent depends on noise more broadly remains open (Section 6).

H2 is not supported in the tested conditions: the Xie-style switch does not improve on continued exact Muon. For H3, the best tested exponent is $p = -0.25$ in every original E3 condition, leaving no variation in the selected exponent to correlate with the noise indicator. Estimated δ at the switch ranges from 0.69 to 4.09. These conditions do not reach the strongly noise-dominated regime $\delta \ll 1$, so the results do not refute the proposed switch in that regime.

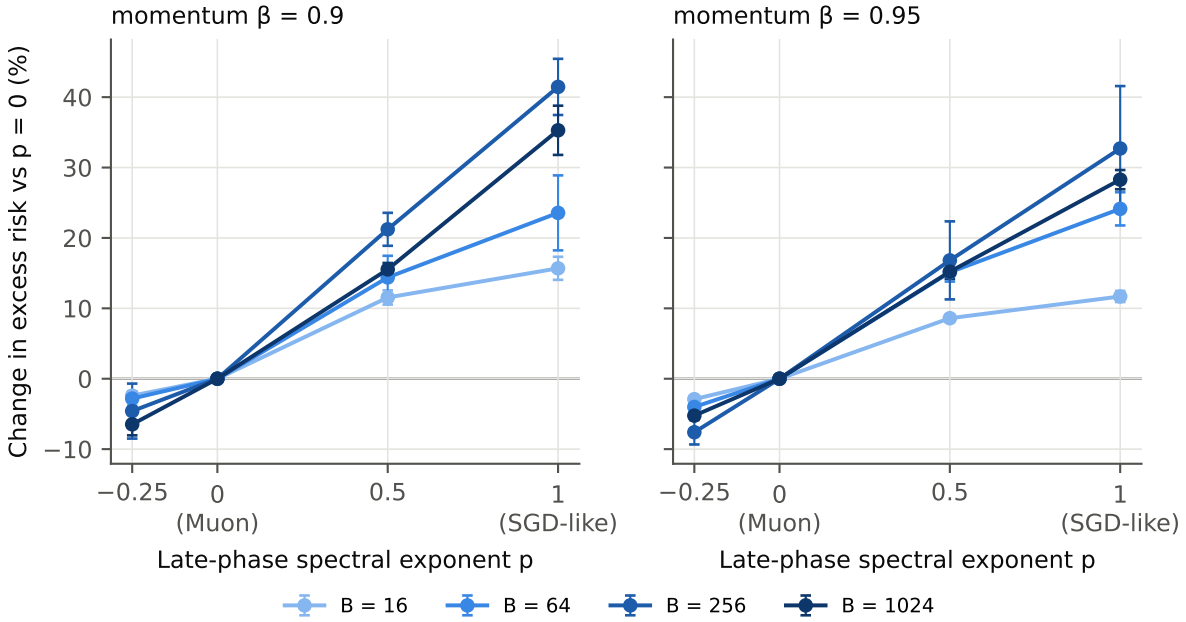


Figure 2: E3 comparison (initially planned, with later tuning amendments). Change in tail excess risk when training continues from the same Muon checkpoint with spectral exponent p instead of Muon ($p = 0$), each arm at its own tuned step size. Percentage change computed within each seed; mean and 95% t-interval over 3 evaluation seeds; below 0 is better than continuing Muon. The $p = -0.25$ arm is best among the original spectral exponents in every condition; the SGD-like end ($p = 1$) has the largest increase in every condition.

6 Exploratory follow-ups

These follow-ups were added after inspecting earlier results and are exploratory. Since -0.25 was at the boundary of the original exponent grid, F1 added $p = -0.5$ and -0.75. Among the tested exponents, -0.25 has the lowest mean risk at $B \leq 64$ and -0.5 at $B \geq 256$, for both momenta (Figure 3). The -0.5 versus -0.25 difference is modest: approximately 1.4–4.1% worse at the smaller batches and 1.3–2.4% better at the larger batches. A scale-free version of the H1 contrast, comparing $\log(L_{p=1}/L_{p=-0.25})$ at $B = 16$ with the same quantity at $B = 1024$, where L is the mean tail excess risk, is -0.20 for $\beta = 0.9$ and -0.16 for $\beta = 0.95$: in H1's direction. It was defined after the results were seen, is exploratory, and does not replace the planned absolute contrast.

An early F1 selection was made before tuning extensions were complete. At the provisional

Table 2: Per condition: Xie's noise indicator δ at the switch; the best exponent among those tested (E3: initially planned with later tuning amendments; F1: exploratory, adds $p = -0.5$ and $p = -0.75$); and the change in tail excess risk against continuing Muon, mean over 3 evaluation seeds.

			Best p		Change vs. Muon (%)		
B	β	δ	E3	F1	$p = -0.25$	$p = 1$	Xie
16	0.9	0.69	−0.25	−0.25	−2.4	+15.7	+15.6
16	0.95	1.10	−0.25	−0.25	−2.9	+11.7	+11.8
64	0.9	1.16	−0.25	−0.25	−2.8	+23.6	+23.5
64	0.95	1.84	−0.25	−0.25	−4.0	+24.1	+23.7
256	0.9	1.77	−0.25	−0.5	−4.6	+41.5	+37.4
256	0.95	2.76	−0.25	−0.5	−7.6	+32.7	+31.2
1024	0.9	2.62	−0.25	−0.5	−6.5	+35.3	+33.2
1024	0.95	4.09	−0.25	−0.5	−5.2	+28.3	+28.3

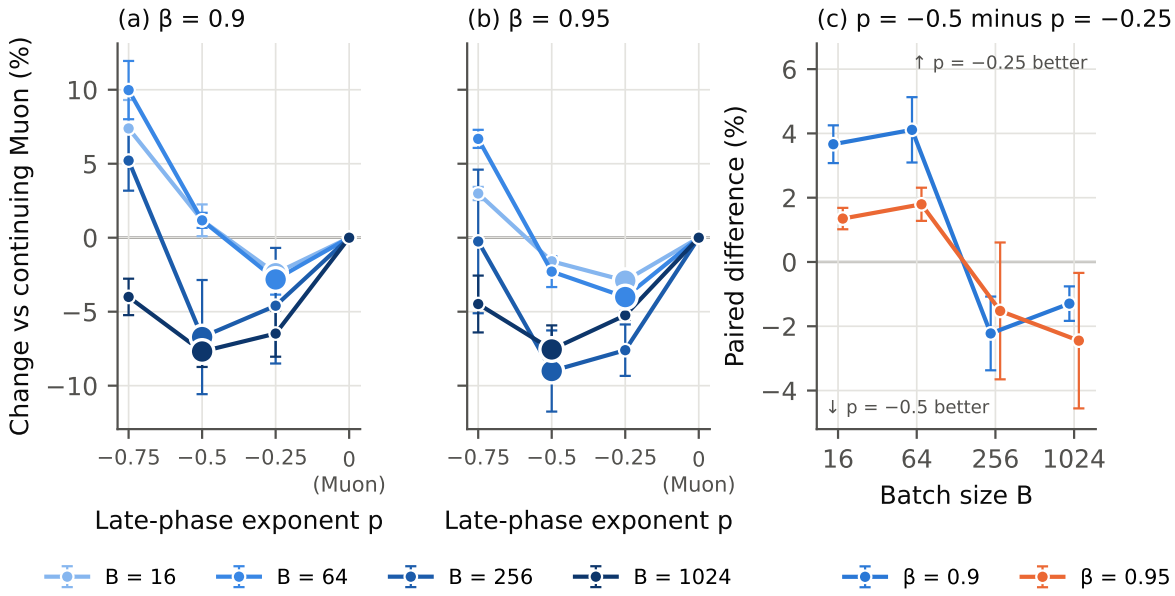


Figure 3: Exploratory follow-up F1 (added after the E3 results). (a, b) As Figure 2, for $p \leq 0$, including the added arms $p = -0.5$ and $p = -0.75$; each batch size's best p is drawn larger. (c) Paired difference between $p = -0.5$ and $p = -0.25$, in % of the mean excess risk at $p = -0.25$; mean and 95% t-interval over 3 seeds. Below 0, flattening more is better; the sign flips between $B = 64$ and $B = 256$.

multiplier $\times 2$, $p = -0.5$ appeared up to 43% worse than Muon in some conditions. Extending the tuning grid and selecting again changed the ranking. This demonstrates that conclusions about spectral shape can depend strongly on the learning-rate search. The scripts now include checks for unfinished tuning and stale selections.

The exploratory frozen control investigates the very small learning rates selected for the SGD-like branches. It holds the two hidden weight matrices fixed while AdamW continues updating the remaining parameters, including all biases. At $B = 16$ and $\beta = 0.9$, the mean tail risks are approximately 0.0672 for $p = 1$ and 0.0675 for the frozen control. Their similar means suggest limited additional improvement from updating these matrices under the selected settings and training budget; they do not establish statistical equivalence or show that the weights themselves remain unchanged. At $B = 1024$ and $\beta = 0.9$, the corresponding risks are approximately 0.0056 and 0.0076, indicating a larger benefit from the hidden-matrix updates.

7 Discussion and limitations

Our experiments support a task-specific benefit from giving greater relative weight to weak singular directions during late-stage training. The $p = -0.25$ branch improves on continued exact Muon across all eight original conditions, while the exploratory extension favours $p = -0.5$ at the larger batch sizes. At the switch, the momentum buffer of each hidden matrix has numerical rank 249–256 out of 256 but an average entropy-based effective rank (the exponential of the entropy of the normalised singular values) of only 33–54 across conditions, so most retained singular directions are weak, and a negative exponent shifts more of the update onto them. These findings are consistent with the qualitative late-stage motivation of Wu et al. (2026), but they do not evaluate the complete DynMuon schedule or establish that negative exponents are preferable on other tasks. For GPT-2, Shumaylov et al. (2026) likewise find the best settings of their update $(GG^\top)^{-c}G$, whose singular-value power $1 - 2c$ is our p, at $c > 1/2$, that is, at negative powers.

The role of gradient noise remains unresolved. Batch size changes both gradient noise and the number of examples processed at a fixed step budget. Different batch sizes and momentum settings also produce different checkpoints and training trajectories. Consequently, the eight conditions do not isolate a causal effect of noise on the preferred exponent.

One possible explanation for the limited small-batch benefit of SGD-like hidden-matrix updates is ill-conditioning: a learning rate small enough for strong directions may produce little progress in weak directions. However, the frozen control measures the benefit of updating these matrices, not the cause of that benefit. Testing this explanation would require additional evidence about curvature or progress along different directions.

The evidence is limited to one synthetic task, one fixed teacher, a small MLP, constant learning rates, and zero weight decay. E1 reuses its tuning seeds for reporting, while the branch comparisons use only three evaluation seeds. The exponent grid is coarse, and F1 was chosen after inspecting E3 results. Timing estimates are specific to CPU float32 computation and exclude data generation and evaluation; learning rates were not retuned for the equal-time budget. The tuning amendments and exploratory analyses further limit confirmatory interpretation.

The most direct next checks are to evaluate the selected E1 configurations on fresh seeds and assess the key branch comparisons on additional seeds. A stronger investigation of the noise mechanism would vary noise from common starting checkpoints while controlling other settings and measuring the resulting noise levels. Further work could target $\delta \ll 1$, test sensitivity to the singular-value cutoff, and examine whether the findings persist under learning-rate decay or on another task.

8 Conclusion

On this synthetic teacher–student task, tuned Muon configurations achieved lower excess risk per training step than SGD and AdamW, but not consistently per estimated CPU compute time. In paired late-stage branches, $p = -0.25$ improved on continued exact Muon across all original conditions. An exploratory extension favoured $p = -0.5$ at larger batch sizes. These findings support further investigation of negative spectral exponents, but do not establish gradient noise as the cause of the observed differences. The tuning history demonstrates that comparisons of spectral update shapes require careful learning-rate selection.

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Appendix

This appendix provides supporting details for the experimental protocol and the interpretation of the measured noise regime. Table 3 records the selected learning-rate multipliers, while Figure 4 visualises the noise indicator measured at the branching point. These details support reproducibility and the scope of the conclusions.

E3 extension rounds reduced edge cells from 37 to 7 to 3 to 0; F1 reduced them from 16 to 12 to 2 to 0. The control and F1 were exploratory; the tuning and timing amendments are dated in `notes/protocol.md`. Scripts, configurations, summaries and reproduction instructions are in the repository.

Table 3: Selected step-size multiplier of every arm in every condition, chosen on tuning seeds 0 and 1. It scales the learning rate at the switch (exact Muon's tuned E1 rate) or, for Xie's switch, Xie's matched momentum-SGD rate.

		Spectral exponent p							
B	β	-0.75	-0.5	-0.25	0	0.5	1	NS5	Xie
16	0.9	4	4	2	1	1/4	1/32	1	1/16
16	0.95	4	2	1	1/2	1/8	1/32	1/2	1/32
64	0.9	16	8	8	2	1	1/2	4	1
64	0.95	16	8	4	2	1/2	1/4	2	1/2
256	0.9	32	16	8	2	2	2	4	2
256	0.95	16	8	4	2	1/2	1/2	2	1
1024	0.9	32	16	4	2	1	1	2	2
1024	0.95	16	8	2	1	1/2	1/2	1	1

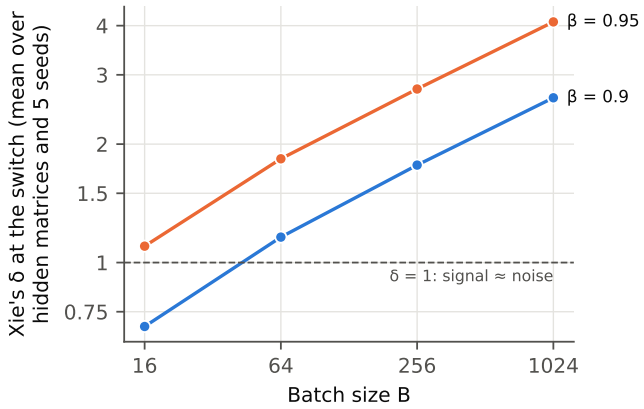


Figure 4: Xie's noise indicator δ at the switch (step 3000), mean over both hidden matrices and 5 seeds. Below 1, the estimated noise in the momentum exceeds the estimated signal, under the assumptions stated in Section 3.